The documentation and process conversion measures necessary to comply with this document shall be completed by 19 April 2024.

INCH-POUND

MIL-PRF-19500/290R
w/AMENDMENT 2
19 January 2024
SUPERSEDING
MIL-PRF-19500/290R
w/AMENDMENT 1
24 August 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, PNP, SILICON, SWITCHING, TYPES, 2N2904, 2N2904A, 2N2904AL, 2N2905, 2N2905A, AND 2N2905AL, JAN, JANTX, JANTXV, JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for PNP, silicon, switching transistors. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500. RHA level designators "M", "D", "P", "L", "R", "F", "G" and "H" are appended to the device prefix to identify devices which have passed RHA requirements.

1.2 Package outlines. The device package outline for this specification sheet is similar to TO-39 in accordance with figure 1 for all encapsulated device types.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T = (1) T_A = +25°C	P_T = (2) T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{STG} and T_J	$R_{\theta JA}$ (1) (2)	$R_{\theta JC}$ (1) (2)
	W	W	V dc	V dc	V dc	mA dc	°C	°C/W	°C/W
2N2904 (1)	0.8	3.0	-60	-40	-5	-600	-65 to +200	195	50
2N2905 (1)	0.8	3.0	-60	-40	-5	-600	-65 to +200	195	50
2N2904A, (2)	0.8	3.0	-60	-60	-5	-600	-65 to +200	195	50
2N2905A, (2)	0.8	3.0	-60	-60	-5	-600	-65 to +200	195	50
2N2904AL, (2)	0.8	3.0	-60	-60	-5	-600	-65 to +200	195	50
2N2905AL (2)	0.8	3.0	-60	-60	-5	-600	-65 to +200	195	50

(1) For derating, see figure 2 and figure 3.
(2) For thermal impedance, see figure 4 and figure 5.

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1.4 Primary electrical characteristics at T_A = +25°C.

	h_{FE} at V_{CE} = -10 V dc				
	h_{FE1} I_C = -0.1 mA dc	h_{FE2} I_C = -1 mA dc	h_{FE3} I_C = -10 mA dc	h_{FE4} (1) I_C = -150 mA dc	h_{FE5} (1) I_C = -500 mA dc
	2N2904	2N2904	2N2904	2N2904	2N2904
Min	20	25	35	40	20
Max		175		120	
	2N2905	2N2905	2N2905	2N2905	2N2905
Min	35	50	75	100	30
Max		450		300	
	2N2904A 2N2904AL	2N2904A 2N2904AL	2N2904A 2N2904AL	2N2904A 2N2904AL	2N2904A 2N2904AL
Min	40	40	40	40	40
Max		175		120	
	2N2905A 2N2905AL	2N2905A 2N2905AL	2N2905A 2N2905AL	2N2905A 2N2905AL	2N2905A 2N2905AL
Min	75	100	100	100	50
Max		450		300	

(1) Pulsed (see 4.5.1).

Types	$\|h_{fe}\|$	C_{obo}	Switching	Switching
2N2904 2N2904A 2N2904AL 2N2905 2N2905A 2N2905AL	f = 100 MHz V_{CE} = -20 V dc I_C = -50 mA dc	100 kHz ≤ f ≤ 1 MHz V_{CB} = -10 V dc I_E = 0	t_{on} (see figure 6)	t_{off} (see figure 7)
		pF	ns	ns
Min	2.0			
Max		8	45	300

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "P", "L", "R", "F", "G", and "H".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

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1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "2904" and "2905".

1.5.4 Suffix symbols. The following suffix letters are incorporated in the PIN in the order listed in the table as applicable:

A	Indicates a modified version of the non-suffix device.
L	For L suffix devices, dimension LL = 1.5 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max.

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

* 1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

* 1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

For device type 2N2904, 2N2904A, 2N2904AL: ..... 300 krads(Si) 1/

For device type 2N2905, 2N2905A, 2N2905AL: ..... 300 krads(Si) 1/

1/ The manufacturers supplying these device types have performed characterization testing in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 300 krads(Si).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

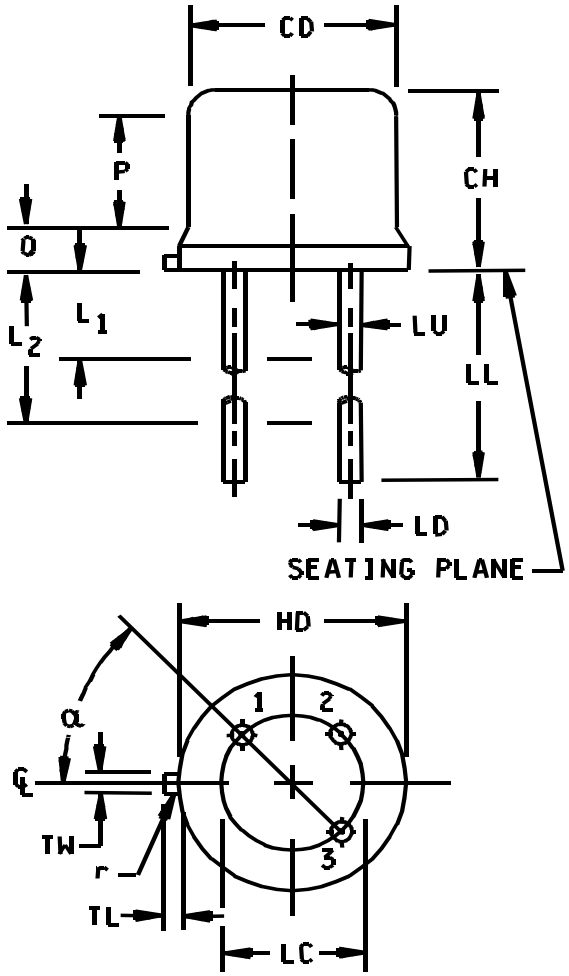
MIL-STD-750 – Test Methods For Semiconductor Devices.

* Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	in	in	mm	mm	Note
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.019	0.41	0.48	7, 8
LL	.500	.750	12.70	19.05	7, 8, 11
LU	.016	.019	0.41	0.48	7, 8
L_1		.050		1.27	7, 8
L_2	.250		6.35		7, 8
P	.100		2.54		
Q		.050		1.27	5
TL	.029	.045	0.74	1.14	4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45° TP		45° TP		6



NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
* 7. Dimension LU applies between L_1 and L2. Dimension LD applies between L2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
* 11. For "L" suffix devices, dimension LL is 1.50 (38.10 mm) minimum, 1.75 (44.45 mm) maximum.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

* FIGURE 1. Physical dimensions (similar to TO-39).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$R_{\theta JA}$ Thermal resistance junction to ambient.
$R_{\theta JC}$ Thermal resistance junction to case.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I, table II, and table III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS level	Measurements for JANTXV and JANTX level
1b	Required	Required (JANTXV only)
3b (1) 3c	Not applicable Required (see 4.3.2)	Not applicable Required (see 4.3.2)
5	Required	Not required
8	Required	Not required
9	I_{CBO2}, h_{FE4}	Not applicable
10	24 hours minimum	24 hours minimum
11	I_{CBO2}; h_{FE4}; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE4} = ±15 percent.	I_{CBO2};h_{FE4}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE4} = ±15 percent.	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE4} = ±15 percent.

(1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements. Record thermal impedance for JANS level devices in screen 13 or anytime after serialization.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = -10 - -30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum, T_A ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, an alternate burn-in flow may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence or adequacy is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The thermal impedance limit used in screen 3c of 4.3 and the subgroup 2 of table I herein shall comply with the thermal impedance graph in figures 4 and 5 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of subgroups 1 and 2, of table I herein, inspection only (table E-VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) of MIL-PRF-19500 and 4.4.2.1 herein. Delta requirements shall be in accordance with 4.5.4 herein: delta requirements only apply to subgroups B4, and B5. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with 4.5.4 herein.

4.4.2.1 Group B inspection (JANS), table E-VIA of MIL-PRF-19500.

*

Subgroup	Method	Condition
B4	1037	V_{CB} = -10 V dc.
B5	1027	V_{CB} = -10 V dc; P_D ≥ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with table E-VIA of MIL-PRF-19500, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve a T_J = +225°C minimum.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action.

*

Subgroup	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = -10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage without exceeding max ratedV_{CEO}, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, TA = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder, or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

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* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Delta requirements shall be in accordance with 4.5.4 herein; delta requirements only apply to subgroup C6.

4.4.3.1 Group C inspection (JANS), table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and applied thermal impedance curves.
C6	1026	1,000 hours at V_{CB} = -10 V dc; power shall be applied to achieve T_J = +150°C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3 n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Group C inspection (JAN, JANTX, and JANTXV), table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3).
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Quality conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750 for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Delta requirements shall be in accordance with4.5.4 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

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* 4.5.3 Displacement damage characterization. For RHA devices, each supplier shall perform a displacement damage characterization. The characterization shall demonstrate exposure versus data and does not indicate pass or fail criteria. The exposure shall be conducted in accordance with method 1017 of MIL-STD-750. The following details shall apply:

a. Samples may be taken from any wafer of the qualification lot.

b. As a minimum, testing shall be at 2E+12 n/cm^2 plus two additional fluence levels chosen by the manufacturer.

c. If the device degrades less than 5 percent of the specification at the highest fluence level, a single data point may be sufficient.

Alternate package options may be substituted for characterization. The displacement damage characterization data shall be made available from the manufacturer.

* 4.5.4 Delta measurements. Delta measurements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit
		Method	Conditions		
1.	Collector-base cutoff current	3036	Bias condition D, V_{CB} = -50 V dc	ΔI_{CB02} (1)	100 percent of initial value or ±8 nA dc, whichever is greater.

(1) Devices which exceed the table I limits for this test shall not be accepted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071					
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Salt atmosphere (corrosion)	1041	n = 6 devices, c = 0, (For laser marked devices only)				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 4/ 6/	1071	n = 22 devices, c = 0				
Fine leak						
Gross leak						
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hours or T_A = +300°C at t = 2 hours n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance 7/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Collector to base cutoff current	3036	Bias condition D, V_{CB} = -60 V dc	I_{CBO1}		-10	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = -5 V dc	I_{EBO1}		-10	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = -10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N2904, 2N2905				-40		V dc
2N2904A, 2N2905A				-60		V dc
2N2904AL, 2N2905AL				-60		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = -50 V dc	I_{CBO2}			
2N2904, 2N2905					-20	nA dc
2N2904A, 2N2905A					-10	nA dc
2N2904AL, 2N2905AL					-10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = -3.5 V dc	I_{EBO2}		-50	nA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter cutoff current	3041	Bias condition C	I_{CES}			
2N2904, 2N2905		V_{CE} = -40 V dc			1	µA dc
2N2904A, 2N2904AL		V_{CE} = -60 V dc			1	µA dc
2N2905A, 2N2905AL		V_{CE} = -60 V dc			1	µA dc
Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -0.1 mA dc	h_{FE1}			
2N2904				20		
2N2905				35		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				75		
Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -1.0 mA dc	h_{FE2}			
2N2904				25	175	
2N2905				50	450	
2N2904A, 2N2904AL				40	175	
2N2905A, 2N2905AL				100	450	
Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -10 mA dc	h_{FE3}			
2N2904				35		
2N2905				75		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				100		
Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -150 mA dc, pulsed (see 4.5.1)	h_{FE4}			
2N2904				40	120	
2N2904A, 2N2904AL				40	120	
2N2905				100	300	
2N2905A, 2N2905AL				100	300	
Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -500 mA dc, pulsed (see 4.5.1)	h_{FE5}			
2N2904				20		
2N2905				30		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				50		
Collector to emitter voltage (saturated)	3071	I_C = -150 mA dc, I_B = -15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		-0.4	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter voltage (saturated)	3071	I_C = -500 mA dc, I_B = -50 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		-1.6	V dc
Base emitter voltage (saturated)	3066	Test condition A, I_C = -150 mA dc, I_B = -15 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		-1.3	V dc
Base emitter voltage (saturated)	3066	Test condition A, I_C = -500 mA dc, I_B = -50 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		-2.6	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D, V_{CB} = -50 V dc	I_{CBO3}			
2N2904, 2N2905					-20	µA dc
2N2904A, 2N2904AL					-10	µA dc
2N2905A, 2N2905AL					-10	µA dc
Low temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -1.0 mA dc	h_{FE6}			
2N2904				15		
2N2905				30		
2N2904A, 2N2904AL				20		
2N2905A, 2N2905AL				50		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -10 V dc, I_C = -1 mA dc, f = 1 kHz	h_{fe}			
2N2904				25		
2N2905				50		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				100		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	Method	Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 4 – Continued.						
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = -20 V dc, I_C =- 50 mA dc, f = 100 MHz	$\|h_{fe}\|$	2.0		
Open circuit output capacitance	3236	V_{CB} = -10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Input capacitance (output open-circuited)	3240	V_{EB} = -2.0 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz (see 4.5.2)	C_{ibo}		30	pF
Turn-on time		(See figure 6)	t_{on}		45	ns
Turn-off time		(See figure 7)	t_{off}		300	ns
Subgroups 5 and 6						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ This hermetic seal test is an end-point to temp-cycling in addition to electrical measurements.
7/ For end-point measurements this test is required for the following subgroups.
Group B, steps 2 and 3 (JAN, JANTX, and JANTXV).
Group B, subgroups 3, 4, and 5 (JANS).
Group C, subgroups 2 and 6.
Group E, subgroups 1 and 2.

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* TABLE II. Group D inspection and end-point limits.

	Inspection 1/ 2/	MIL-STD-750		Symbol	Limits	Limits	Unit
		Method	Conditions		Min	Max	
	Subgroup 1						
	Neutron irradiation	1017					
	Collector to base cutoff current	3036	Bias condition D, V_{CB} = -60 V dc	I_{CBO1}		-20	µA dc
	Collector to base cutoff current	3036	Bias condition D	I_{CBO2}			
	2N2904, 2N2905		V_{CB} = -50 V dc			-40	nA dc
	2N2904A, 2N2905A					-20	nA dc
	2N2904AL, 2N2905AL					-20	nA dc
*	Forward current transfer ratio	3076	V_{CE} = -10 V dc; IC = -0.1 mA dc	[h_{FE1}]			
	2N2904				[10]		
	2N2905				[17.5]		
	2N2904A, 2N2904AL				[20]		
	2N2905A, 2N2905AL				[37.5]		
*	Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -1.0 mA dc	[h_{FE2}]			
	2N2904				[12.5]	175	
	2N2905				[25]	450	
	2N2904A, 2N2904AL				[20]	175	
	2N2905A, 2N2905AL				[50]	450	
*	Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -10 mA dc, pulsed (see 4.5.1)	[h_{FE3}]			
	2N2904				[17.5]		
	2N2905				[37.5]		
	2N2904A, 2N2904AL				[20]		
	2N2905A, 2N2905AL				[50]		

See footnotes at end of table.

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1 - Continued						
* Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -150 mA dc	[h_{FE4}]			
2N2904				[20]	120	
2N2905				[50]	300	
2N2904A, 2N2904AL				[20]	120	
2N2905A, 2N2905AL				[50]	300	
* Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C =- 500 mA dc	[h_{FE5}]			
2N2904				[10]		
2N2905				[15]		
2N2904A, 2N2904AL				[20]		
2N2905A, 2N2905AL				[25]		
* Collector to emitter voltage (saturated)	3071	I_C = -150 mA dc, I_B = -15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$	[		
2N2904					-0.46	V dc
2N2905					-0.46	V dc
2N2904A, 2N2904AL					-0.46	V dc
2N2905A, 2N2905AL					-0.46	V dc

See footnotes at end of table

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1 – Continued						
* Collector to emitter voltage (saturated)	3071	I_C = -500 mA dc, I_B = -50 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$			
2N2904					-1.84	V dc
2N2905					-1.84	V dc
2N2904A, 2N2904AL					-1.84	V dc
2N2905A, 2N2905AL					-1.84	V dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = -5 V dc	I_{EBO1}		-20	µA
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = -3.5 V dc	I_{EBO2}		-100	Na
Base to emitter voltage (saturated)	3066	Test condition A, I_C = -150 mA dc, I_B = -15 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		-1.5	V dc
Base to emitter voltage (saturated)	3066	Test condition A, I_C = -500 mA dc, I_B = -50 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		-3.0	V dc
Breakdown voltage, Collector to emitter	3011	Bias condition D; I_C = -10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N2904, 2N2905				-40		V dc
2N2904A, 2N2905A				-60		V dc
2N2904AL, 2N2905AL				-60		V dc
Collector to emitter cutoff current	3041	Bias condition C	I_{CES}			
2N2904, 2N2905		V_{CE} = -40 V dc			-2.0	µA
2N2904A, 2N2905A		V_{CE} = -60 V dc			-2.0	µA
2N2904AL, 2N2905AL		V_{CE} = -60 V dc			-2.0	µA

See footnotes at end of table

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2						
Steady-state total dose Irradiation	1019					
2N2904, 2N2905 2N2904A, 2N2905A 2N2904AL, 2N2905AL		Gamma exposure V_{CES} = -32 V Gamma exposure V_{CES} = -48 V Gamma exposure V_{CES} = -48 V				
Collector to base cutoff current	3036	Bias condition D, V_{CB} = -60 V dc	I_{CBO1}		-20	
Collector to base cutoff current	3036	Bias condition D V_{CB} = -50 V dc	I_{CBO2}			µA
2N2904, 2N2905 2N2904A, 2N2905A 2N2904AL, 2N2905AL					-40 -20 -20	nA nA nA
Forward current transfer ratio 2N2904 2N2905 2N2904A, 2N2904AL 2N2905A, 2N2905AL	3076	V_{CE} = -10 V dc, I_C = -0.1 mA dc	[h_{FE1}]	 [10] [17.5] [20] [37.5]		
Forward current transfer ratio 2N2904 2N2905 2N2904A, 2N2904AL 2N2905A, 2N2905AL	3076	V_{CE} = -10 V dc, I_C = -1 mA dc	[h_{FE2}]	 [12.5] [25] [20] [50]		
Forward current transfer ratio 2N2904 2N2905 2N2904A, 2N2904AL 2N2905A, 2N2905AL	3076	V_{CE} = -10 V dc, I_C = -10 mA dc, pulsed (see 4.5.1)	[h_{FE3}]	 [17.5] [37.5] [20] [50]		
Forward current transfer ratio 2N2904 2N2905 2N2904A, 2N2904AL 2N2905A, 2N2905AL	3076	V_{CE} = -10 V dc, I_C = -150 mA dc	[h_{FE4}]	 [20] [50] [20] [50]		

See footnotes at end of table

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued.						
* Forward current transfer ratio	3076	V_{CE} = -10 V dc, I_C = -500 mA dc	[h_{FE5}]			
2N2904				[10]		
2N2905				[15]		
2N2904A, 2N2904AL				[20]		
2N2905A, 2N2905AL				[25]		
* Collector to emitter voltage (saturated)	3071	I_C = -150 mA dc, I_B = -15 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)\,1}$			
2N2904					-0.46	V dc
2N2905					-0.46	V dc
2N2904A, 2N2904AL					-0.46	V dc
2N2905A, 2N2905AL					-0.46	V dc
* Collector to emitter voltage (saturated)	3071	I_C = -500 mA dc, I_B = -50 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)\,2}$			
2N2904					-1.84	V dc
2N2905					-1.84	V dc
2N2904A, 2N2904AL					-1.84	V dc
2N2905A, 2N2905AL					-1.84	V dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = -5 V dc	I_{EBO1}		-20	µA
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = -3.5 V dc	I_{EBO2}		-100	nA
Base to emitter voltage (saturated)	3066	Test condition A, I_C = -150 mA dc, I_B = -15 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		-1.5	V dc
Base to emitter voltage (saturated)	3066	Test condition A, I_C = -500 mA dc, I_B = -50 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		-3.0	V dc

See footnotes at end of table.

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued.						
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N2904, 2N2905				-40		V dc
2N2904A, 2N2905A				-60		V dc
2N2904AL, 2N2905AL				-60		V dc
Collector to emitter cutoff current	3041	Bias condition C	I_{CES}			
2N2904, 2N2905		V_{CE} = -40 V dc			-2.0	µA
2N2904A, 2N2905A		V_{CE} = -60 V dc			-2.0	µA
2N2904AL, 2N2905AL		V_{CE} = -60 V dc			-2.0	µA

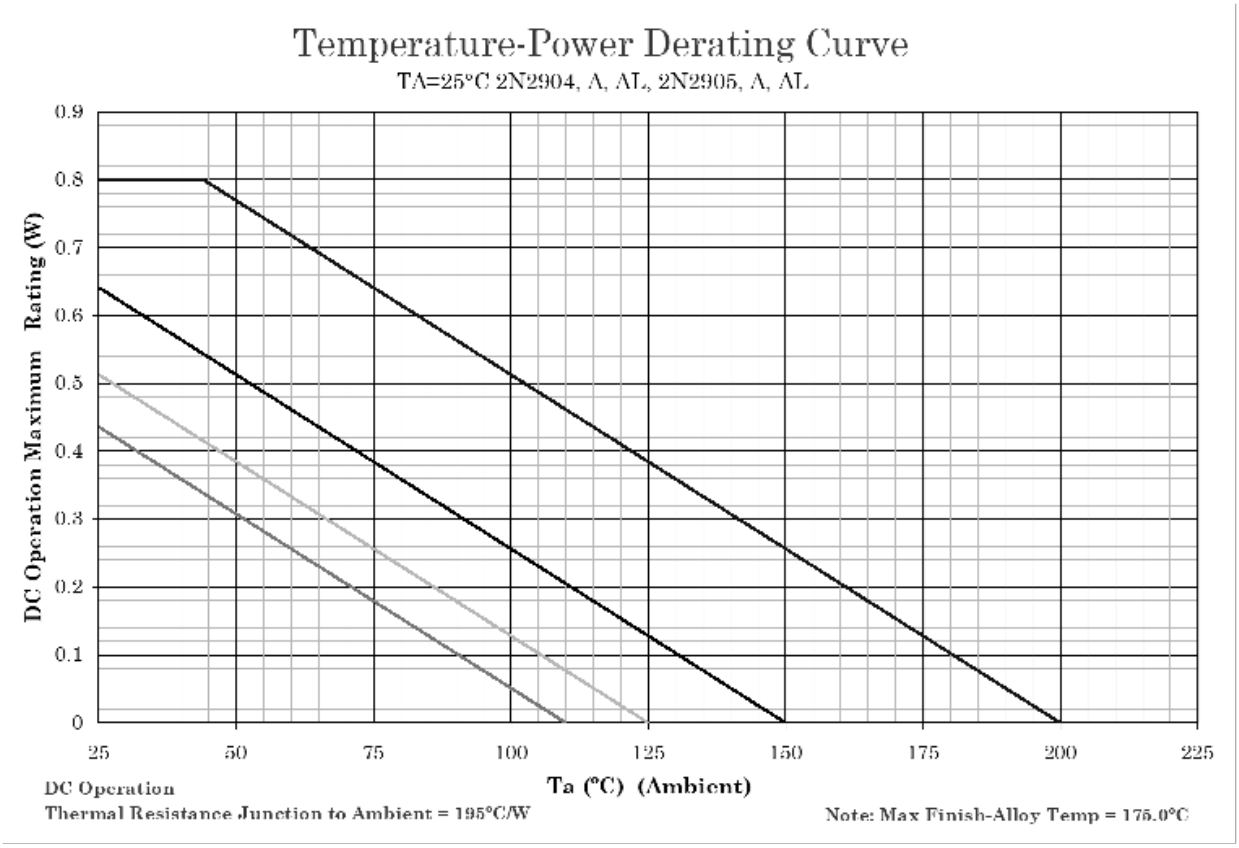
1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to the corresponding A and AL suffix versions unless otherwise noted.
* 4/ See 6.2.f herein.
5/ See method 1019 of MIL-STD-750 for how to determine [h_{FE}] by first calculating the delta ($1/h_{FE}$) from the pre- and post-radiation h_{FE}. Notice the [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and 4.5.4 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = -10 V dc, 6,000 cycles.	
Electrical measurements		See table I, subgroup 2 and 4.5.4 herein.	
Subgroup 4			
Thermal impedance curves		See table E-IX of MIL-PRF-19500, Group E, subgroup 4.	
Subgroup 5			
Not applicable			
Subgroup 6			11 devices
Electrostatic discharge (ESD)	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	
* Subgroup 12			
Neutron Irradiation	1017	See 4.5.3	

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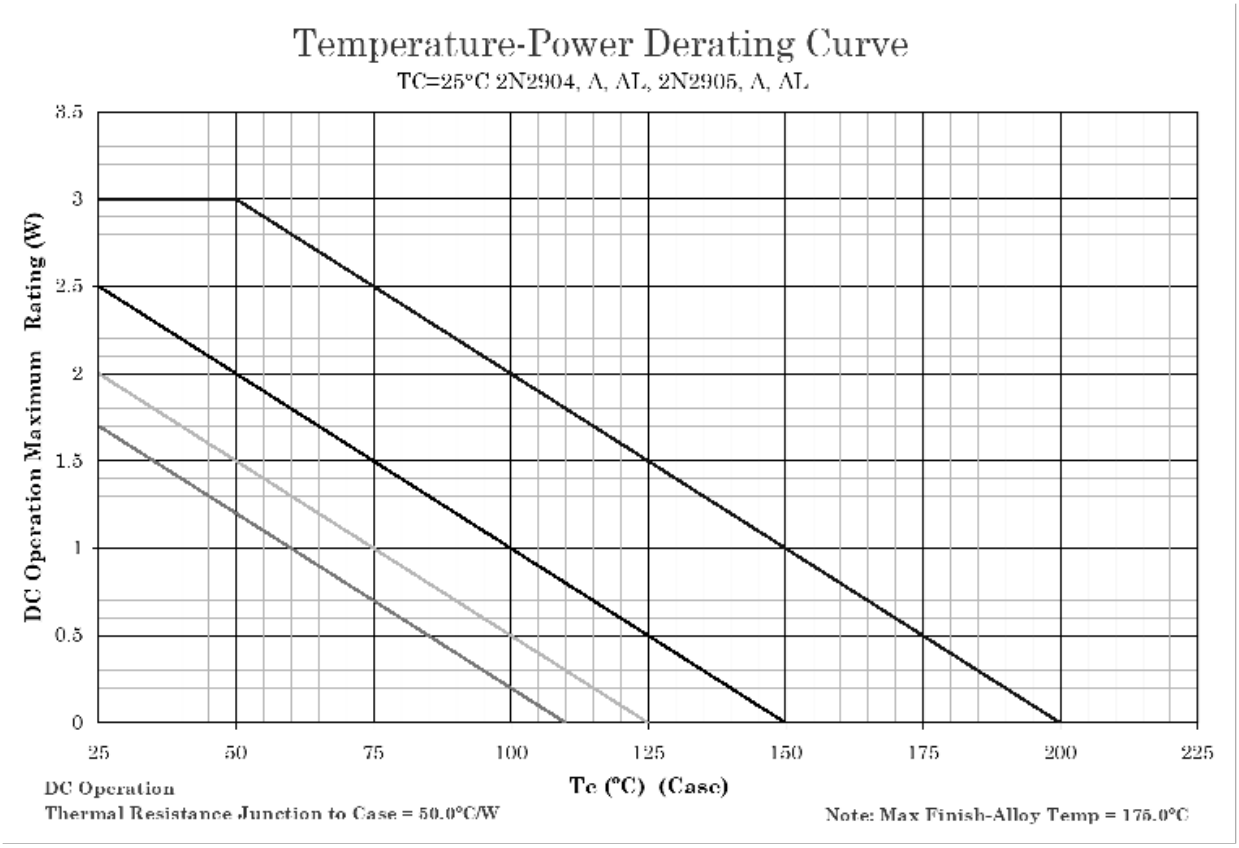


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 2. Derating for 2N2904, 2N2904A, 2N2904AL, 2N2905, 2N2905A, 2N2905AL, ($R_{\theta JA}$) PCB (TO-39).

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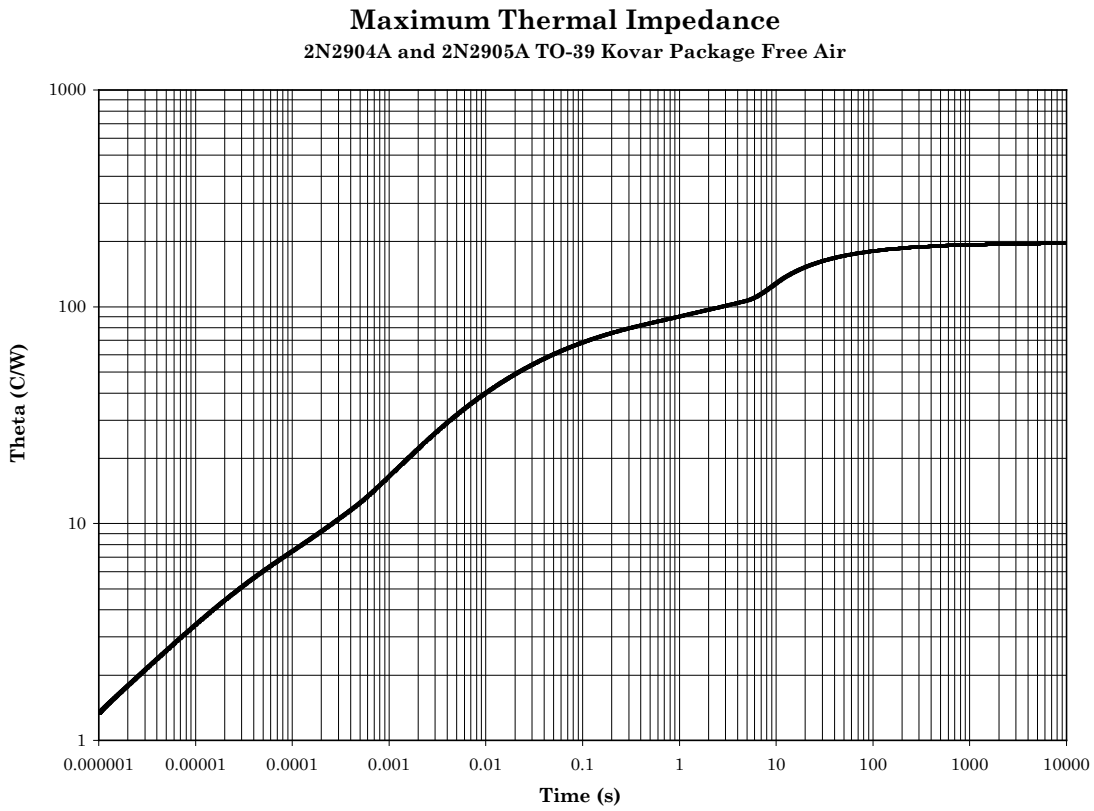


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (TJ ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 3. Derating for 2N2904, 2N2904A, 2N2904AL, 2N2905, 2N2905A, 2N2905AL, ($R_{\theta JC}$) PCB (TO-39).

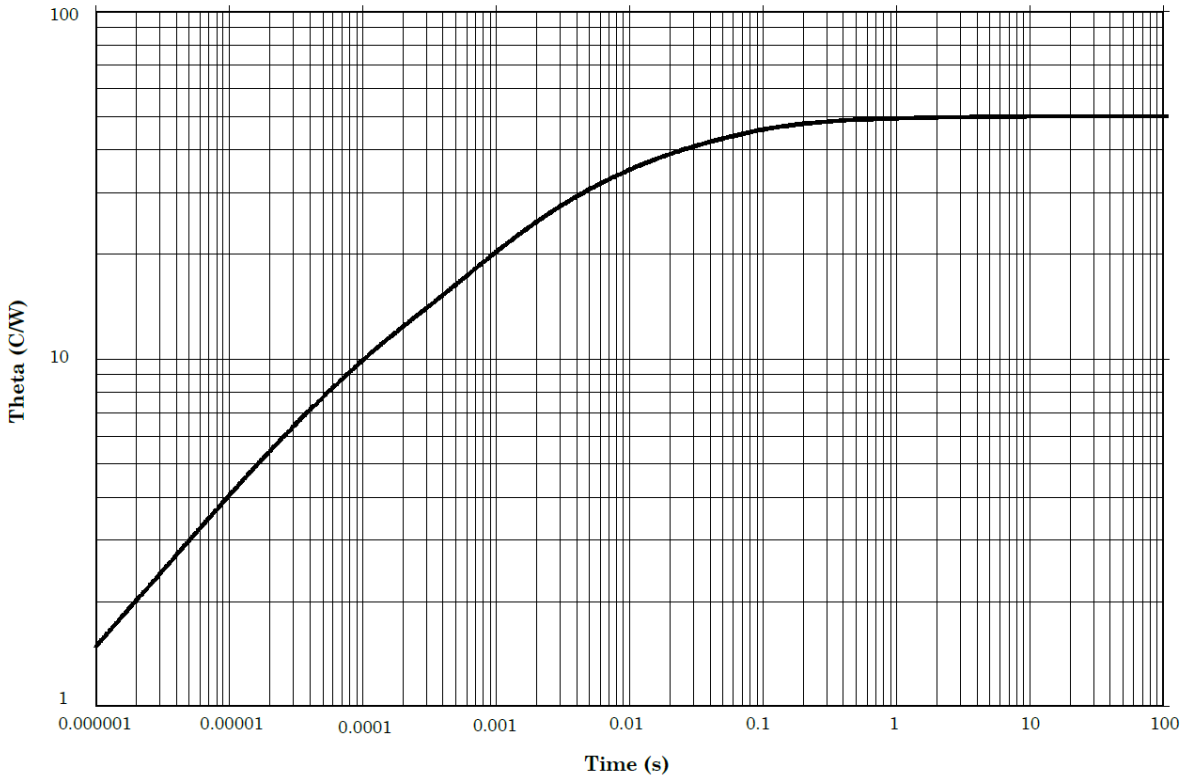
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Ambient free air cooled T_A = +25°C, 800mW, thermal resistance $R_{\theta JA}$ = 195°C/W.

FIGURE 4. Thermal impedance graph ($R_{\theta JA}$) for all 2N2904, and 2N2905 devices (TO-39).

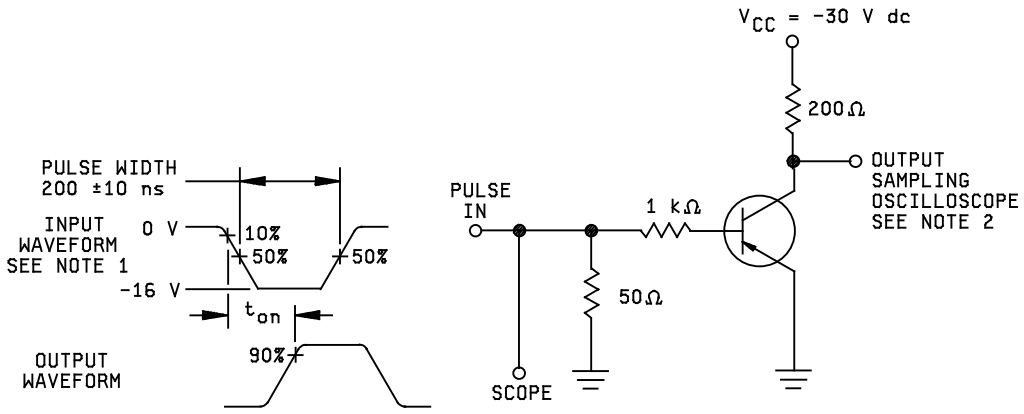
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Ambient case mounted T_C = +25°C, thermal resistance $R_{\theta JC}$ = 50°C/W.

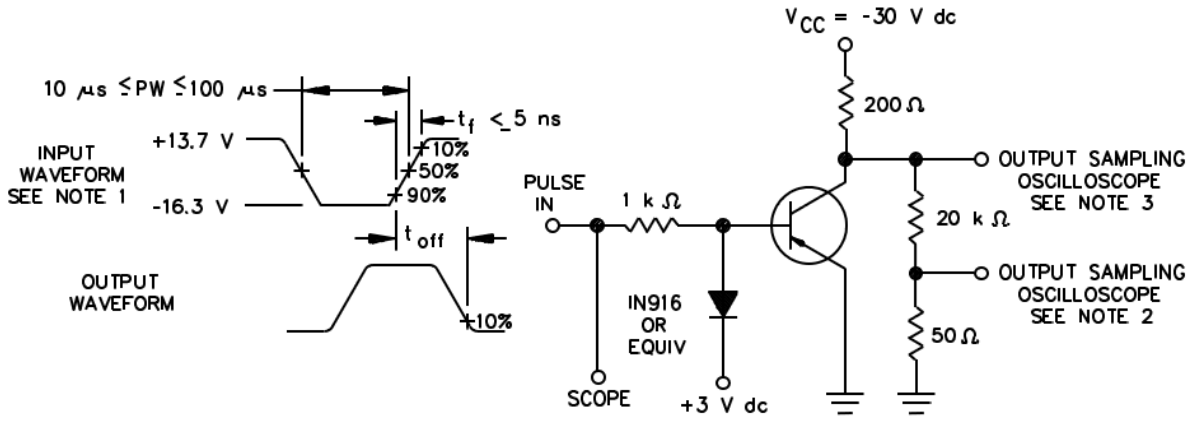
FIGURE 5. Thermal impedance graph ($R_{\theta JC}$) for all 2N2904, and 2N2905 devices (TO-39).

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 6. Saturated turn-on switching time test circuit.



NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.
3. Alternate test point for high impedance attenuating probe.

FIGURE 7. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

e. The complete PIN, see 1.5 and 6.4.

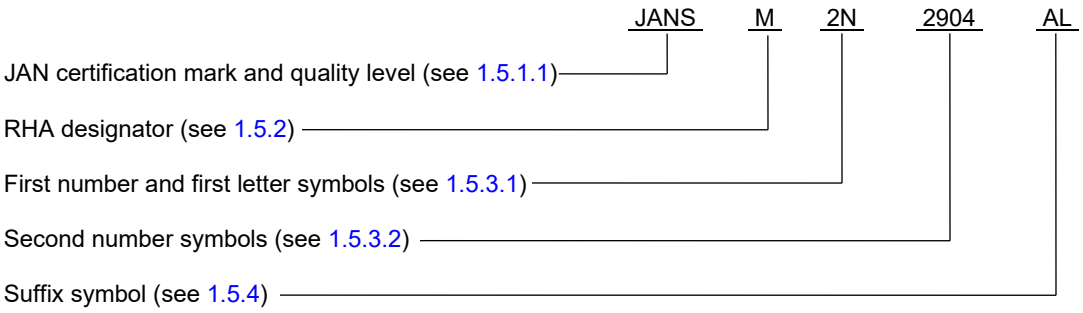
f. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it shall be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil . An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for type 2N2904 and 2N2905.			
JAN2N2904	JANTX2N2904	JANTXV#2N2904	JANS#2N2904
JAN2N2904A	JANTX2N2904A	JANTXV#2N2904A	JANS#2N2904A
JAN2N2904AL	JANTX2N2904AL	JANTXV#2N2904AL	JANS#2N2904AL
JAN2N2905	JANTX2N2905	JANTXV#2N2905	JANS#2N2905
JAN2N2905A	JANTX2N2905A	JANTXV#2N2905A	JANS#2N2905A
JAN2N2905AL	JANTX2N2905AL	JANTXV#2N2905AL	JANS#2N2905AL

(1) The number sign (#) represent one of eight RHA designators available (M, D, P, L, R, F, G, or H). The PIN is also available without a RHA designator.

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6.6 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
* NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2023-101)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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